

OFFICIAL ANSWER KEY & EVALUATION SCHEME

Subject: Advanced Data Structures and Algorithms | Paper: Midterm Examination (Backtracking Optimized)

Q#	Topic / Unit	Type / Marks	Correct Answer / Solution Key
Q1	Divide and conquer design paradigms (Unit 1)	Short Answer (2M)	Detailed analytical solution based on Divide and conquer design paradigms curriculum specifications.
Q2	Greedy choice property (Unit 2)	Short Answer (2M)	Detailed analytical solution based on Greedy choice property curriculum specifications.
Q3	Elements of dynamic programming (Unit 3)	Short Answer (2M)	Detailed analytical solution based on Elements of dynamic programming curriculum specifications.
Q4	Backtracking paradigm (Unit 4)	Short Answer (2M)	Detailed analytical solution based on Backtracking paradigm curriculum specifications.
Q5	NP-Hard (Unit 5)	Short Answer (2M)	Detailed analytical solution based on NP-Hard curriculum specifications.
Q6	Divide and conquer design paradigms (Unit 1)	Long Answer (10M)	Detailed analytical solution based on Divide and conquer design paradigms curriculum specifications.
Q6	Divide and conquer design paradigms (Unit 1)	Short Answer (10M)	Detailed analytical solution based on Divide and conquer design paradigms curriculum specifications.
Q7	Greedy choice property (Unit 2)	Long Answer (10M)	Detailed analytical solution based on Greedy choice property curriculum specifications.
Q7	Greedy choice property (Unit 2)	Short Answer (10M)	Detailed analytical solution based on Greedy choice property curriculum specifications.
Q8	Elements of dynamic programming (Unit 3)	Long Answer (10M)	Detailed analytical solution based on Elements of dynamic programming curriculum specifications.
Q8	Elements of dynamic programming (Unit 3)	Short Answer (10M)	Detailed analytical solution based on Elements of dynamic programming curriculum specifications.
Q9	Backtracking paradigm (Unit 4)	Long Answer (10M)	Detailed analytical solution based on Backtracking paradigm curriculum specifications.
Q9	Backtracking paradigm (Unit 4)	Short Answer (10M)	Detailed analytical solution based on Backtracking paradigm curriculum specifications.